# COMPUTER PROGRAMMING PROJECT

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## 1 Introduction

**ABSTRACT**

Surface blast design calculations are often time-consuming and susceptible to errors arising from incorrect formula selection, inaccurate data entry, and unit conversion mistakes. The BlastSmart project addresses these challenges by developing specialised calculators for surface blast design and Kuz–Ram fragmentation analysis as part of a mobile application for blasting engineers. The calculators are designed to automate blast design calculations and improve the accuracy and efficiency of the blasting process. Throughout the calculations, a coupling ratio of 100% is assumed, meaning that the explosive column completely fills the blast hole diameter.

The BlastSmart calculator provides an effective computational tool for surface blast design and fragmentation analysis. However, site-specific calibration is required before implementation in field operations to ensure that the calculator accurately reflects the geological and operational conditions of a particular mine.

**BACKGROUND**

Dwala Global Resources aims to develop a mobile application, BlastSmart, to assist blasting engineers by automating blast design calculations and reducing the reliance on manual computations and paper-based methods.

The BlastSmart application enables blasting engineers to calculate, analyse, and compare blast design parameters directly from a mobile device. Phase 1 of the project focuses on the development of two core computational modules: the Surface Blast Design Calculator and the Kuz–Ram Fragmentation Calculator.

The Surface Blast Design Calculator requires the following input parameters, hole diameter, explosive density,bench height, spacing to burden ratio,Technical powder factor, relative effective explosive energy(REE), rock type, distance from private property

The actual powder factor equation varies according to the rock type. For hard-rock conditions, the subdrill is included in the charged length, whereas for soft-rock conditions, the subdrill is subtracted from the charged length. Throughout the calculations, a coupling ratio of 100% is assumed, implying that the explosive column completely occupies the blast hole diameter.

**LITERATURE REVIEW** The linear explosive mass is calculated from the density and cross-sectional area of a fully coupled explosive column.The standard relationship between hole diameter, explosive density, charge concentration, stemming, burden, and subdrilling. The burden equation used in the calculator comes from the powder-factor relationship. The rock volume assigned to one hole is $B \times S \times BH$, and $S=rB$. Stemming is often related to the burden or the hole diameter. We used a stemming value of around 0.7 times the burden while field rules often use 20 to 30 times the hole diameter. Subdrilling is also often expressed as a fraction of the burden.In this project we used 0.25 times the burden. The average fragment size is calculated using the Kuz–Ram model. This model came from Kuznetsovs work on energy charge mass, powder factor and average fragment size back, in 1973. The Kuz–Ram model was further developed by Cunningham in 1983 and 1987. He took Kuznetsovs relationship and turned it into the Kuz–Ram model.

**CALCULATOR DESIGN**

The aim of this calculator design is to create an easy-to-use specialized blasting calculator for surface mines that will enable efficient computation and validation of blast designs in real time. To achieve these results, a range of operation variables have been integrated with the underlying algorithmic logic into a single computational tool.

The calculator requires the following inputs: · Bench height (BH) · In hole explosives density(d) · Burden ratio(S/B) · Relative Effective explosive energy(E) · Rock factor (A) The calculator outputs the following: · Stemming Length (SL) · Burden(B) · Actual powder factor ($PF_{act}$ ) · Mean fragmentation size ($X_{50}$ )

It calculates the linear explosive mass using:

$M= dD^2/1273$

The program then checks the distance from private property. A stemming length of 30BD is used when the distance is greater than 1,000 m, while 20BD is used when the distance is smaller.

After calculating stemming, the calculator determines burden, spacing, and sub-drill:

B=[(M(BHSL))/((S/B)PFBH)]

For hard rock:

PFact=(M x (BH-SL+SD))/(B x S x BH)

For soft rock:

PFact=(M x BH-SL-SD)/ ($B \times S \times BH$)

The fragmentation function calculates the charge quantity:

Q=M x (BH-SL)

It then uses the actual powder factor, rock factor, relative explosive energy, and charge quantity to calculate X50

**DISCUSSION**

The higher powder factor for rock results in a smaller burden and spacing. This means each hole has rock but more explosive per cubic metre. As a result the actual powder factor is higher. The predicted fragment size is slightly smaller.

Stemming helps control gas confinement. More stemming reduces issues like collar ejection, airblast and flyrock. However much stemming causes poor fragmentation near the top of the bench. A stemming length of 6 m is. Needs to be checked against actual performance.

The predicted fragmentation has an X of 47–54 cm. This means half of the material may be larger than this size. Therefore it's essential to compare predicted fragmentation with the loader bucket and crusher capacity to anticipate material.

There are three issues that need confirmation:

The BD unit must be in millimetres for the linear-mass equation but in metres for stemming. The algorithm applies more stemming from the property. This is opposite to the expected need for confinement near sensitive areas. The pseudocodes energy exponent is 0.633. This differs from the published Kuz–value of 19/20 as mentioned by Cunningham in 1987. The model assumes the rock is uniform. It may underestimate fines. Factors, like joints, water, drilling deviation and initiation timing can cause fragmentation to differ from predictions as noted by Cunningham in 2005.

**Conclusion and Future Work**

We did the part of the project and it worked out well. We made a calculator that figures out the way to blast the surface. This calculator is connected to another calculator that helps us understand how the rocks will break apart. The system calculates a lot of things, like how explosive to use how to set it up and how big the pieces of rock will be.

For the part of the project we should work on a few more things. We need to make a calculator that can predict how fast the rocks will fly when we blast them. We also need to make a version of the calculator for underground blasting.. We should make it so the calculator can do the math really fast. The calculator should also be able to change units check to make sure the input is correct and be adjusted to work for mines. It would be great if we could also save the records of blasts and see graphs that show how all the rocks were broken apart. We should add all these things to the surface blast calculator and the Kuz-Ram mean-fragmentation calculator

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